

BSC106

General Physics

Mechanics 1

1.1 Introduction

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1.1 Introduction: Units, dimensions, and measurements

- Any number that is used to describe a physical phenomenon quantitatively is called a **physical quantity**
- Any physical quantity has a **dimension** associated with it e.g. length, time, and mass
- if x is a quantity of length and measured result is ten meters, we write $x = 10 \text{ m}$ where the **unit** is meter
- if t is a quantity of time and is measured to be five seconds, so $t = 5 \text{ s}$ where the **unit** is second
- Can $x + t = 15$?
- Can x/t ?

- Adding and subtracting quantities of different dimension does not make sense.
- But multiplying and dividing them, we obtain new types of quantities, which we call **derived quantities**

$$\frac{x}{t} = \frac{10 \text{ m}}{5 \text{ s}} = 2 \text{ m s}^{-1} = 2 \text{ m/s}.$$

- **speed** is the ratio of a length to a time

- Standardize unit for measured quantities
- International System of Units, abbreviated SI/metric system
- m, kg, s for length, mass and time

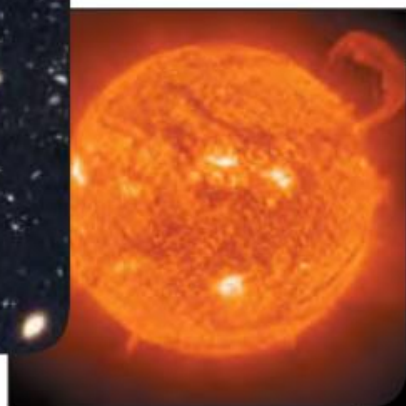
Table 1.1 Some Units of Length, Mass, and Time

Length	Mass	Time
1 nanometer = 1 nm = 10^{-9} m (a few times the size of the largest atom)	1 microgram = 1 μ g = 10^{-6} g = 10^{-9} kg (mass of a very small dust particle)	1 nanosecond = 1 ns = 10^{-9} s (time for light to travel 0.3 m)
1 micrometer = 1 μ m = 10^{-6} m (size of some bacteria and living cells)	1 milligram = 1 mg = 10^{-3} g = 10^{-6} kg (mass of a grain of salt)	1 microsecond = 1 μ s = 10^{-6} s (time for space station to move 8 mm)
1 millimeter = 1 mm = 10^{-3} m (diameter of the point of a ballpoint pen)	1 gram = 1 g = 10^{-3} kg (mass of a paper clip)	1 millisecond = 1 ms = 10^{-3} s (time for sound to travel 0.35 m)
1 centimeter = 1 cm = 10^{-2} m (diameter of your little finger)		
1 kilometer = 1 km = 10^3 m (a 10-minute walk)		

1.5 Some typical lengths in the universe. (f) is a scanning tunneling microscope image of atoms on a crystal surface; (g) is an artist's impression.



(a) 10^{26} m
Limit of the
observable
universe



(b) 10^{11} m
Distance to
the sun



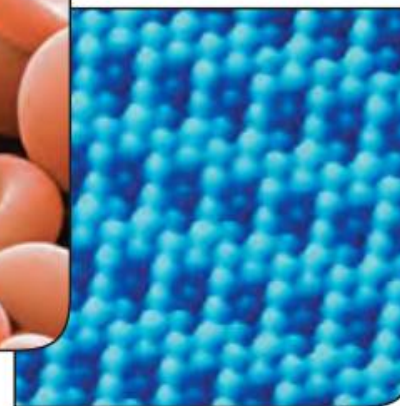
(c) 10^7 m
Diameter of
the earth



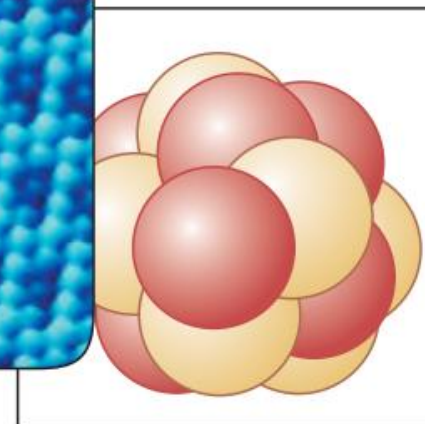
(d) 1 m
Human
dimensions



(e) 10^{-5} m
Diameter of a
red blood cell



(f) 10^{-10} m
Radius of an
atom



(g) 10^{-14} m
Radius of an
atomic nucleus

- SI unit for power, called the watt (W), is defined in terms of the base units for mass, length, and time.

$$1 \text{ watt} = 1 \text{ W} = 1 \text{ kg} \cdot \text{m}^2/\text{s}^3$$

- Changing/conversion of unit

$$2 \text{ min} = (2 \text{ min})(1) = (2 \cancel{\text{ min}}) \left(\frac{60 \text{ s}}{1 \cancel{\text{ min}}} \right) = 120 \text{ s.}$$

- Order of magnitude
- Prefixes can be used with any base units as multipliers. For example, for the metre (m) as the base unit

$$1 \text{ mm} = 10^{-3} \text{ m (milimetre),}$$

$$1 \text{ km} = 10^3 \text{ m (kilometre).}$$

Power	Prefix	Abbreviation
10^{-24}	yocto	y
10^{-21}	zepto	z
10^{-18}	atto	a
10^{-15}	femto	f
10^{-12}	pico	p
10^{-9}	nano	n
10^{-6}	micro	μ
10^{-3}	mili	m
10^{-2}	centi	c
10^{-1}	deci	d
10^3	kilo	k
10^6	mega	M
10^9	giga	G
10^{12}	tera	T
10^{15}	peta	P
10^{18}	exa	E
10^{21}	zetta	Z
10^{24}	yotta	Y

Dimensional analysis

- We will denote the **dimensions** of mass, length, and time by M , L , and T , respectively.
- If x is known to represent a distance/length, then we say the dimension of x is L .

$$[x] = L$$

$$\begin{aligned} A &= \text{width} \times \text{height} \\ &= xy. \end{aligned}$$

$$[A] = [x][y] = L \cdot L = L^2.$$

- A standard SI unit of a particular physical quantity is easily obtained by replacing L with the SI unit meter (m), M by the SI unit for mass (kg), and T with the SI unit second (s).
- For instance, average velocity is given by

$$v = \frac{\text{displacement}}{\text{time}} = \frac{s}{t}.$$

$$[v] = \frac{[s]}{[t]} = \frac{L}{T} = LT^{-1}$$

- Therefore the SI unit for average velocity is

$$LT^{-1} \rightarrow \text{ms}^{-1} = \text{m/s}$$

Example 1.1.3:

An object starting from rest at time $t = 0$ with constant acceleration a has a displacement given by

$$s = \frac{1}{2}at^2.$$

- (a) Determine the dimensions of acceleration.
- (b) If $s = 5.00$ m, $t = 30.0$ s, determine the value of a with the correct SI units.

a) Acceleration is defined as **change in velocity per unit time**.

Velocity has dimensions: $[LT^{-1}]$

Divide by time: $[LT^{-2}]$

So:

$$[a] = LT^{-2}$$

b) Solve for a

From the displacement equation:

$$s = \frac{1}{2}at^2 \quad \text{and} \quad a = \frac{2s}{t^2}$$

$$a = \frac{2(5.00)}{(30.0)^2}$$

$$a = \frac{10.0}{900}$$

$$a = 0.0111 \text{ m/s}^2$$

Example 1.1.4:

The mass of a solid cube is 856 g, and each edge has a length of 5.35 cm. Determine the density ρ of the cube in SI units.

volume of cube

$$V = L^3 = (0.0535 \text{ m})^3$$
$$V = 1.53 \times 10^{-4} \text{ m}^3$$

Density formula

$$\rho = \frac{m}{V} = \frac{0.856}{1.53 \times 10^{-4}}$$

$$\rho \approx 5.59 \times 10^3 \text{ kg/m}^3$$

1.2 Kinematics in 1 Dimension

Moves along a single axis: one-dimensional motion

- The motion is along a straight line only: vertical, horizontal, or slanted, but it must be straight.
- Not considering any force yet (Mechanics 2, week 3)
- The moving object is either a particle (by which we mean a point-like object such as an electron) or an object that moves like a particle

1.2.1 Displacement and velocity

- A moving object will have different **position** at different times. If we let x be distance, and t be the time, we can view x as a function of t

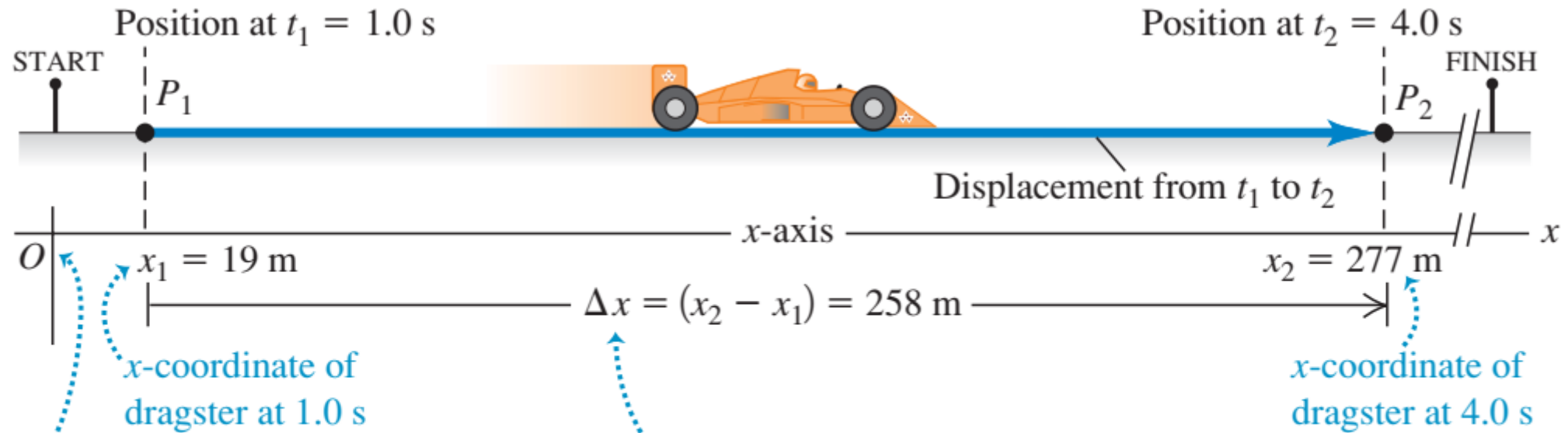
We define the **displacement** of an object between time t_1 and t_2 as

$$s = \Delta x = x(t_2) - x(t_1), \quad t_2 \geq t_1.$$

A **time interval** is a difference in time,

$$\Delta t = t_2 - t_1, \quad t_2 \geq t_1.$$

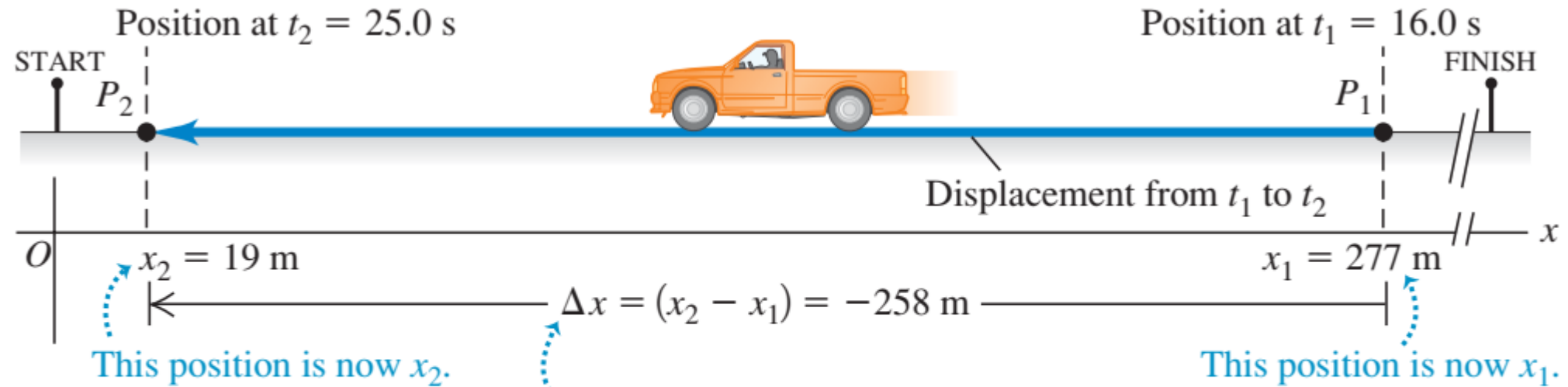
$$v_{\text{av}} = \frac{\Delta x}{\Delta t} = \frac{x(t_2) - x(t_1)}{t_2 - t_1}, \quad t_2 > t_1. \quad [v_{\text{av}}] = \frac{[x(t_2) - x(t_1)]}{[t_2 - t_1]} = \frac{\text{length}}{\text{time}} = \frac{L}{T}.$$



x is positive to the right of the origin (O), negative to the left of it.

When the dragster moves in the $+x$ -direction, the displacement Δx is positive and so is the average x -velocity:

$$v_{\text{av-}x} = \frac{\Delta x}{\Delta t} = \frac{258 \text{ m}}{3.0 \text{ s}} = 86 \text{ m/s}$$



When the truck moves in the $-x$ -direction, Δx is negative and so is the average x -velocity:

$$v_{\text{av-}x} = \frac{\Delta x}{\Delta t} = \frac{-258 \text{ m}}{9.0 \text{ s}} = -29 \text{ m/s}$$

- Instantaneous velocity

$$t_1 = t, \quad t_2 = t + \Delta t. \quad \lim_{\Delta t \rightarrow 0} v_{\text{av}} = \lim_{\Delta t \rightarrow 0} \frac{x(t + \Delta t) - x(t)}{\Delta t}.$$

$$v(t) = \frac{dx}{dt} = \dot{x}(t).$$

- **Speed** is the magnitude of velocity,

$$\text{speed} = |v|$$

Example 1.2.1:

If a person starts driving from Kuala Lumpur at 1:00pm, and arrives at Ipoh at 3:00pm, and the distance between KL and Ipoh is 181.75 km, (see Fig. 1.1) what is the person's average velocity in km h^{-1} and m s^{-1} ?

Average velocity in km/h

$$v_{avg} = \frac{\Delta x}{\Delta t} = \frac{181.75 \text{ km}}{2.00 \text{ h}}$$
$$v_{avg} = 90.875 \text{ km/h}$$

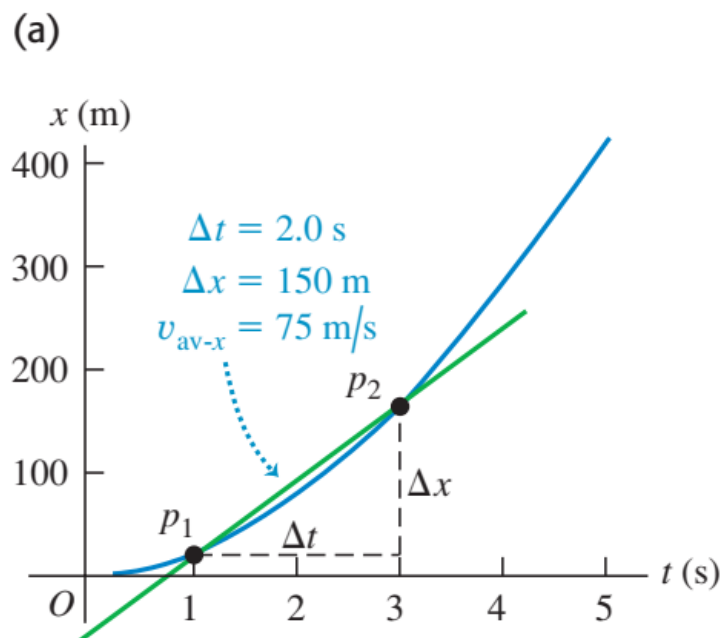
Conversion factor:

$$1 \text{ km/h} = \frac{1000 \text{ m}}{3600 \text{ s}} = 0.2778 \text{ m/s}$$

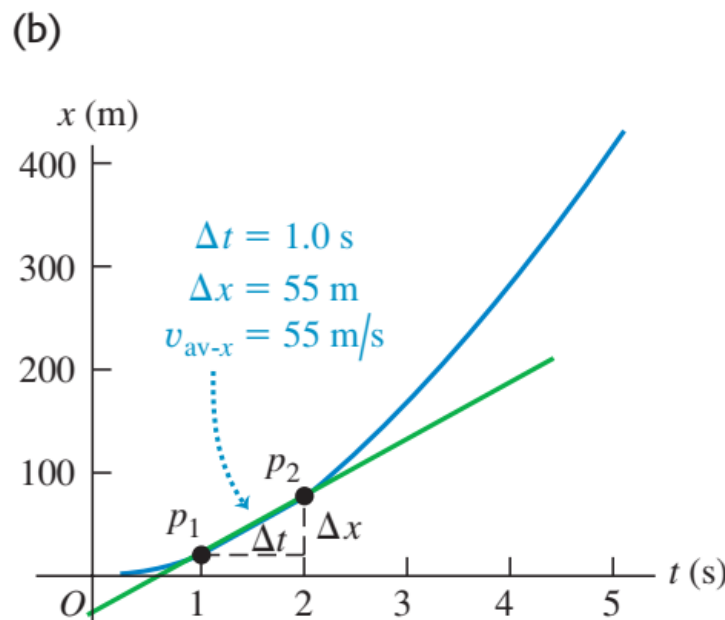
So:

$$v_{avg} = 90.875 \times 0.2778 \approx 25.24 \text{ m/s}$$

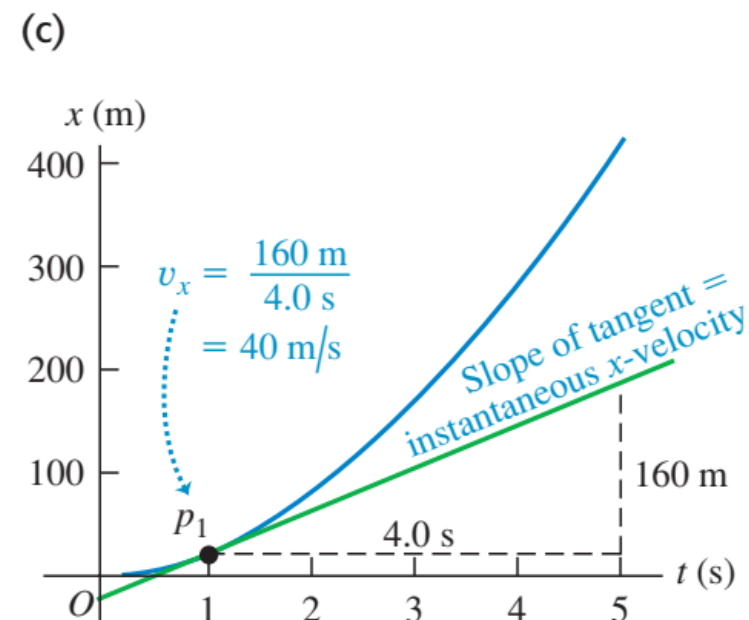
2.7 Using an x - t graph to go from (a), (b) average x -velocity to (c) instantaneous x -velocity v_x . In (c) we find the slope of the tangent to the x - t curve by dividing any vertical interval (with distance units) along the tangent by the corresponding horizontal interval (with time units).



As the average x -velocity v_{av-x} is calculated over shorter and shorter time intervals ...



... its value $v_{av-x} = \Delta x / \Delta t$ approaches the instantaneous x -velocity.



The instantaneous x -velocity v_x at any given point equals the slope of the tangent to the x - t curve at that point.

The **average velocity** is:

$$v_{avg} = \frac{\Delta x}{\Delta t}$$

Graphically: this is the slope of the straight line connecting p_1 and p_2 .

Example 1.2.2:

A car is stopped at a traffic light. It then travels along a straight line, so that its distance from the traffic light is given by

$$x(t) = bt^2 - ct^3, \quad (1.22)$$

where $b = 2.4 \text{ ms}^{-2}$ and $c = 0.12 \text{ m s}^{-3}$.

- (a) Calculate the average velocity of the car for the time interval between $t = 0 \text{ s}$ and $t = 5 \text{ s}$, and also for between $t = 5 \text{ s}$ and $t = 10 \text{ s}$.
- (b) Calculate the instantaneous velocity of the car at $t = 0 \text{ s}$, 5 s , and 10 s .
- (c) How long after $t = 0 \text{ s}$ is the car at rest again?

(b) Instantaneous velocity

$$v(t) = 2bt - 3ct^2$$

- $v(0) = 0$
- $v(5) = 15 \text{ m/s}$
- $v(10) = 12 \text{ m/s}$

$$x(t) = bt^2 - ct^3$$

with

$$b = 2.4 \text{ m/s}^2$$

$$c = 0.12 \text{ m/s}^3$$

(a) Average velocity

$$v_{avg} = \frac{x(t_2) - x(t_1)}{t_2 - t_1}$$

$$\bullet 0 \text{ to } 5 \text{ s: } x(5) = 45 \text{ m, } v_{avg} = 9.0 \text{ m/s}$$

$$\bullet 5 \text{ to } 10 \text{ s: } x(10) = 120 \text{ m, } v_{avg} = 15.0 \text{ m/s}$$

c) Car at rest again

$$2bt - 3ct^2 = 0, t = \frac{2b}{3c} = 13.3 \text{ s}$$

1.2.2 Average and Instantaneous Acceleration

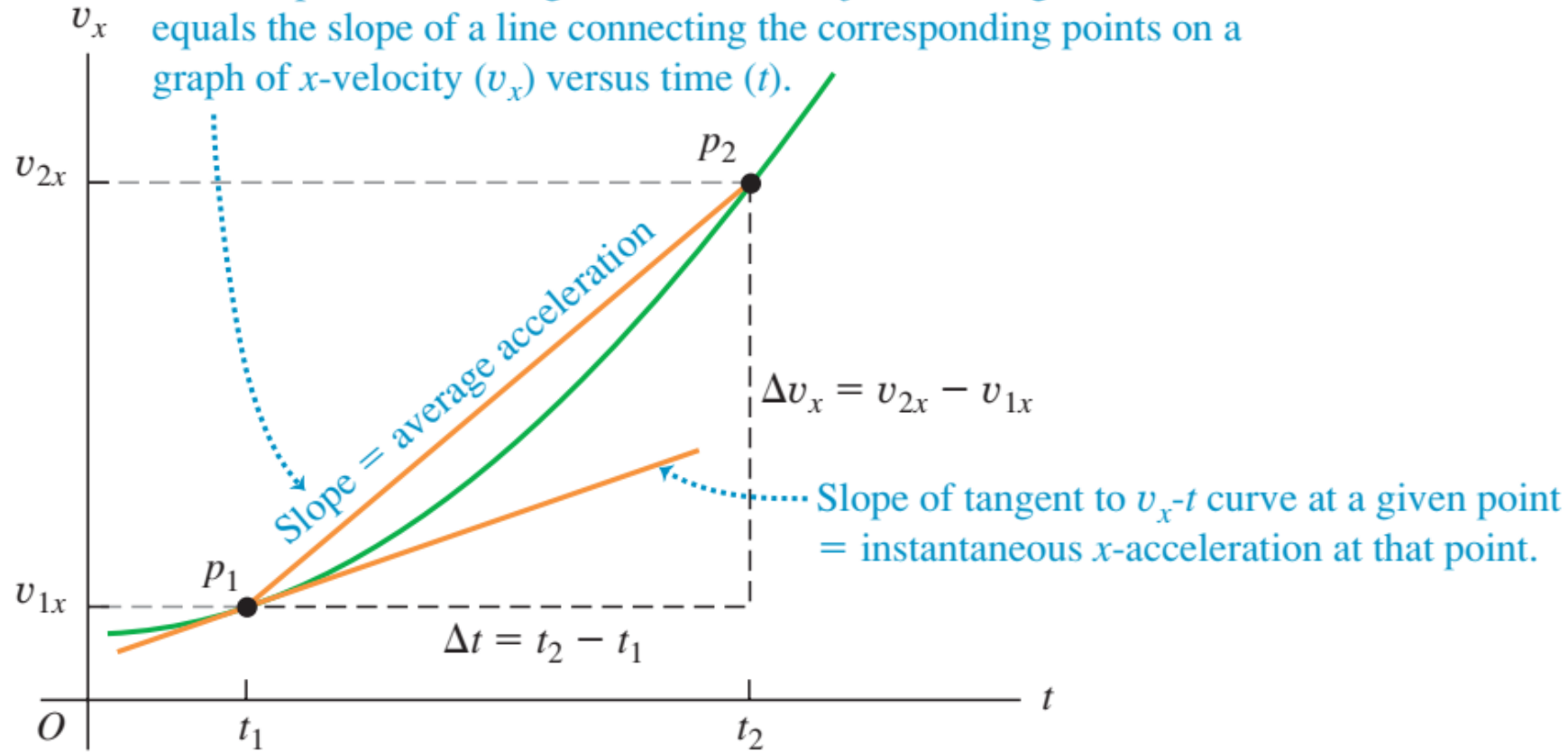
$$a_{\text{av}} = \frac{v(t_2) - v(t_1)}{t_2 - t_1} = \frac{\Delta v}{\Delta t}.$$

$$a = \lim_{\Delta t \rightarrow 0} a_{\text{av}} = \lim_{\Delta t \rightarrow 0} \frac{\Delta v}{\Delta t} = \frac{dv}{dt}.$$

$$a = \frac{dv}{dt} = \frac{d^2x}{dt^2} = \ddot{x}(t).$$

$$[a] = \left[\frac{dv}{dt} \right] = \left[\lim_{\Delta t \rightarrow 0} \frac{\Delta v}{\Delta t} \right] = \lim_{\Delta t \rightarrow 0} \frac{[\Delta v]}{[\Delta t]} = \frac{L/T}{T} = LT^{-2}.$$

For a displacement along the x -axis, an object's average x -acceleration equals the slope of a line connecting the corresponding points on a graph of x -velocity (v_x) versus time (t).



Example 1.2.3:

The velocity of a car is given by

$$v(t) = 60.0 \text{ m s}^{-1} + (0.500 \text{ m s}^{-3}) t^2.$$

- (a) Find the average acceleration in the time interval between $t_1 = 1.00 \text{ s}$ and $t_2 = 3.00 \text{ s}$.
- (b) Derive an expression for the instantaneous acceleration as a function of time.
- (c) Find the instantaneous acceleration at time $t_1 = 1.0 \text{ s}$ and $t_2 = 3.0 \text{ s}$.

(a) Average acceleration

$$a_{avg} = \frac{v(t_2) - v(t_1)}{t_2 - t_1}$$
$$v(1) = 60.5 \text{ m/s}, v(3) = 64.5 \text{ m/s}$$
$$a_{avg} = \frac{64.5 - 60.5}{3 - 1} = 2.0 \text{ m/s}^2$$

(b) Instantaneous acceleration

$$a(t) = \frac{dv}{dt} = (0.500)(2t) = t \text{ (m/s}^2\text{)}$$

(c) Values at specific times

- $a(1) = 1.0 \text{ m/s}^2$
- $a(3) = 3.0 \text{ m/s}^2$

1.2.3 Motion at constant acceleration

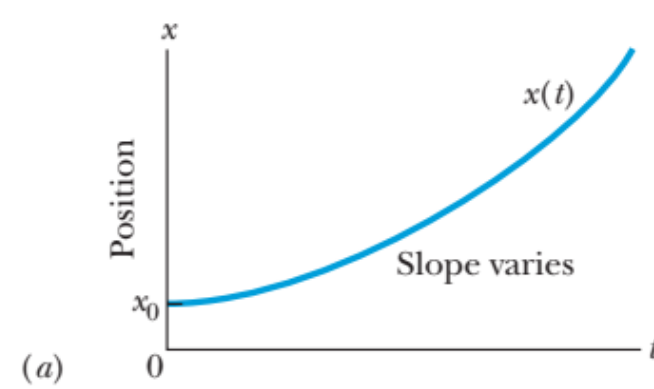
$$\frac{dv}{dt} = a = \text{constant}$$
$$\int_{v_0}^v dv = \int_0^t a \, dt \quad \rightarrow \quad v = v_0 + at,$$

where v_0 is the velocity of the object at $t = 0$.

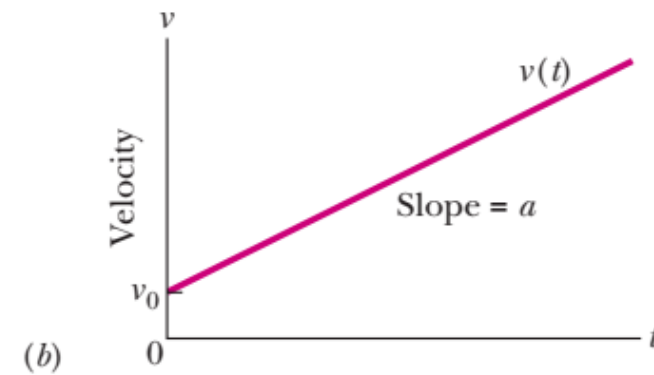
Also, since $\frac{dx}{dt} = v$, we have

$$\int_{x_0}^x dx = \int_0^t v \, dt = \int_0^t (v_0 + at) \, dt$$
$$x = x_0 + v_0 t + \frac{1}{2}at^2,$$

Figure 2-9 (a) The position $x(t)$ of a particle moving with constant acceleration. (b) Its velocity $v(t)$, given at each point by the slope of the curve of $x(t)$. (c) Its (constant) acceleration, equal to the (constant) slope of the curve of $v(t)$.



Slopes of the position graph are plotted on the velocity graph.



Slope of the velocity graph is plotted on the acceleration graph.



Table 2-1 Equations for Motion with Constant Acceleration^a

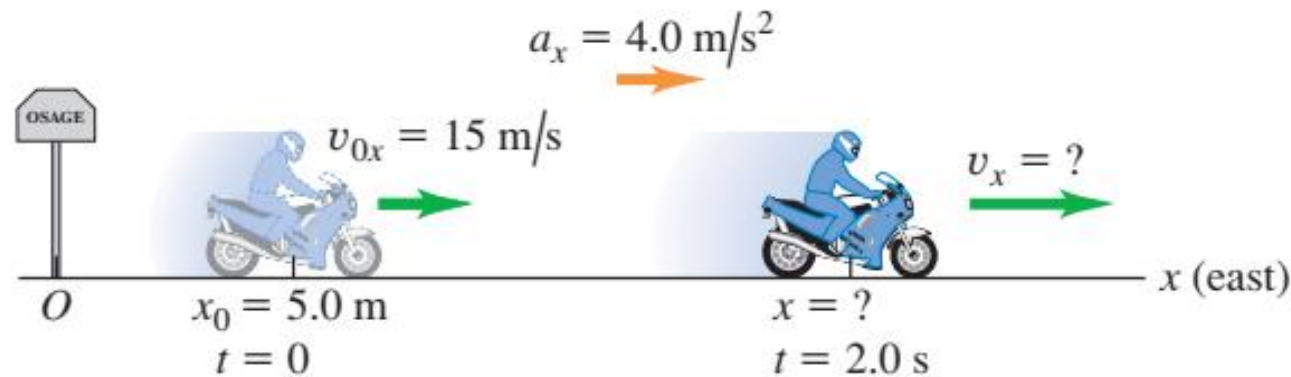
Equation Number	Equation	Missing Quantity
2-11	$v = v_0 + at$	$x - x_0$
2-15	$x - x_0 = v_0t + \frac{1}{2}at^2$	v
2-16	$v^2 = v_0^2 + 2a(x - x_0)$	t
2-17	$x - x_0 = \frac{1}{2}(v_0 + v)t$	a
2-18	$x - x_0 = vt - \frac{1}{2}at^2$	v_0

^aMake sure that the acceleration is indeed constant before using the equations in this table.

Example 2.4 Constant-acceleration calculations

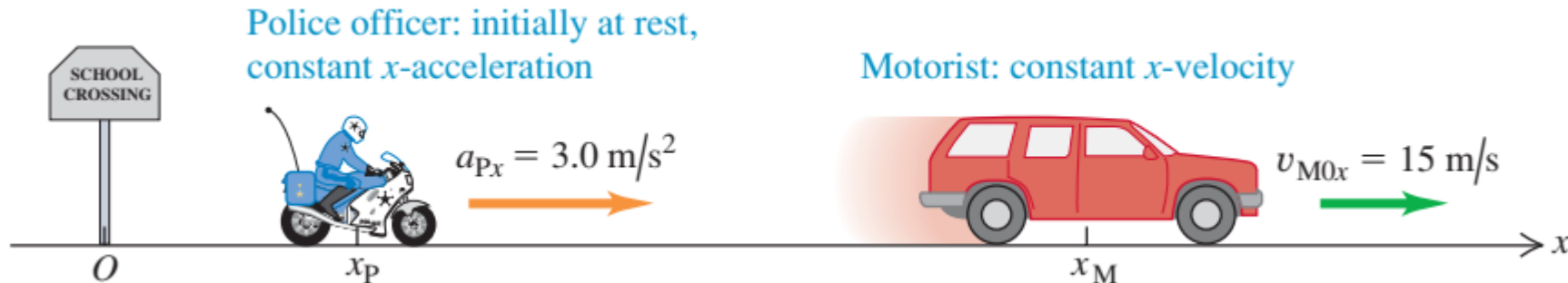
A motorcyclist heading east through a small town accelerates at a constant 4.0 m/s^2 after he leaves the city limits (Fig. 2.20). At time $t = 0$ he is 5.0 m east of the city-limits signpost, moving east at 15 m/s . (a) Find his position and velocity at $t = 2.0 \text{ s}$. (b) Where is he when his velocity is 25 m/s ?

2.20 A motorcyclist traveling with constant acceleration.



- Two bodies with different accelerations

A motorist traveling with a constant speed of 15 m/s (about 34 mi/h) passes a school-crossing corner, where the speed limit is 10 m/s (about 22 mi/h). Just as the motorist passes the school-crossing sign, a police officer on a motorcycle stopped there starts in pursuit with a constant acceleration of 3.0 m/s^2 (Fig. 2.21a). (a) How much time elapses before the officer passes the motorist? (b) What is the officer's speed at that time? (c) At that time, what distance has each vehicle traveled?



Example 2.6 A freely falling coin

A one-euro coin is dropped from the Leaning Tower of Pisa and falls freely from rest. What are its position and velocity after 1.0 s, 2.0 s, and 3.0 s?

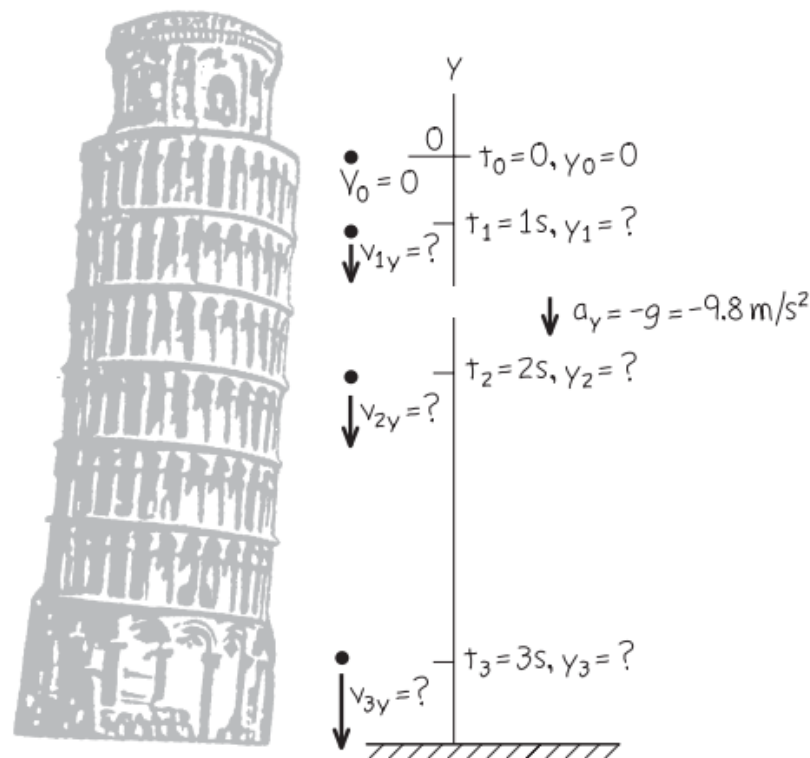
SOLUTION

IDENTIFY and SET UP: “Falls freely” means “falls with constant acceleration due to gravity,” so we can use the constant-acceleration equations. The right side of Fig. 2.23 shows our motion diagram for the coin. The motion is vertical, so we use a vertical

2.23 A coin freely falling from rest.

The Leaning Tower of Pisa

Our sketch for the problem



coordinate axis and call the coordinate y instead of x . We take the origin O at the starting point and the *upward* direction as positive. The initial coordinate y_0 and initial y -velocity v_{0y} are both zero. The y -acceleration is downward (in the negative y -direction), so $a_y = -g = -9.8 \text{ m/s}^2$. (Remember that, by definition, g itself is a positive quantity.) Our target variables are the values of y and v_y at the three given times. To find these, we use Eqs. (2.12) and (2.8) with x replaced by y . Our choice of the upward direction as positive means that all positions and velocities we calculate will be negative.

EXECUTE: At a time t after the coin is dropped, its position and y -velocity are

$$y = y_0 + v_{0y}t + \frac{1}{2}a_y t^2 = 0 + 0 + \frac{1}{2}(-g)t^2 = (-4.9 \text{ m/s}^2)t^2$$

$$v_y = v_{0y} + a_y t = 0 + (-g)t = (-9.8 \text{ m/s}^2)t$$

When $t = 1.0 \text{ s}$, $y = (-4.9 \text{ m/s}^2)(1.0 \text{ s})^2 = -4.9 \text{ m}$ and $v_y = (-9.8 \text{ m/s}^2)(1.0 \text{ s}) = -9.8 \text{ m/s}$; after 1 s, the coin is 4.9 m below the origin (y is negative) and has a downward velocity (v_y is negative) with magnitude 9.8 m/s.

We can find the positions and y -velocities at 2.0 s and 3.0 s in the same way. The results are $y = -20 \text{ m}$ and $v_y = -20 \text{ m/s}$ at $t = 2.0 \text{ s}$, and $y = -44 \text{ m}$ and $v_y = -29 \text{ m/s}$ at $t = 3.0 \text{ s}$.

EVALUATE: All our answers are negative, as we expected. If we had chosen the positive y -axis to point downward, the acceleration would have been $a_y = +g$ and all our answers would have been positive.